

Internship Assignment Tokheim

Reflection

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Inhoudsopgave

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1. Inleiding

This document reflects on my internship experience, covering what went well and what was challenging. I'll walk through the actual results I achieved and evaluate if the final product benefits the company. I'll also dive into the process behind the results, how everything came together, the hurdles I faced, and how I handled them. In short, this reflection shows what I learned on the job and where I still have room to grow.

2. Inhoudelijke reflectie

The internship started out as one simple integration in One Portal. Providing an interactive map and coupling it with the existing alerts module. But over a couple of weeks, the project grew clearer and more defined. Allowing for more interesting extensions and assignments to be added.

Assignments:

- **Tutorial:** To get me warmed up to the codebase, I first had to complete a tutorial. This helped me to acclimatize and get used to all the tools that are used. I mostly worked with the Angular framework writing typescript. Occasionally, I needed to make changes in the SQL database or in the .NET backend.
- **Station Geocoding:** Tokheim has access to station address data for all their clients. My first assignment was to find a way to translate these addresses to usable coordinates to be used in a map
- **Map Study:** Once the coordinates were acquired, the next step was to perform a study on which mapping library most suited our use case. During this stage, I quickly developed a couple prototypes to put next to each other to compare. From this study, Leaflet was chosen as the clear winner.
- **One Portal Map Integration:** The largest portion of my internship was spent on building the map. At the beginning, the assignment was very loosely defined. So, my focus was aimed at getting the basic functionality working first. This meant getting coordinates from the database to create markers and displaying these in the right position. Creating popups for the markers and tying the alerts module to the station markers. From then on, my mentor and I sat down weekly to brainstorm on new ideas and decide on which extensions got priority. From these meetings, I kept a weekly updated list of features to be added.
- **Station Configurator Extension:** Once the map was built, we still had one issue, what to do with newly created stations? My last assignment was to extend the station configurator to automatically provide newly created stations with coordinates. This led to an automatic API call to Nominatim to fetch coordinates when creating a station. Also to correct any misconfigured stations, the forms to update the stations were extended. These now contain a separate map with a drag functionality. The station marker can be dragged and dropped to the right location to set the position of the station.

The fun part, but also the challenge of this internship assignment, was the freedom I got with the execution of the project. The scope was not clearly defined and my mentor entrusted me with a lot of control over the direction and decision making of this project. In the beginning, this surely was a bit of a challenge for me. It was at times hard to determine if I was making the right decisions and making sufficient progress. But by keeping track of everything I do within my notebook, I managed to find a good system. I wrote down as much as I could, because I knew this would benefit me in the long run. This helped me immensely to keep track of my progress. It was also a great way to draw inspiration, any ideas or questions I got, I wrote down. Then later, I could bring it up during stand up or my weekly meeting with my mentor. To me, this was what helped me the most to deal with the open assignment.

2.1. Tutorial

The tutorial was a great way of introducing the codebase. At the end of it, I already got to learn most of the coding guidelines and standards that are used in the team. I learned a lot during this first week, it put me more at ease and it set me up for success in the upcoming weeks.

2.2. Station Geocoding

To translate the station addresses, I saw fit to write a Python converter script. This script took some time to run, about 8 hours, but as it was a one-time conversion, this was a great way to quickly get access to coordinates. I chose to do this in python, because the panda's library was ideal for quickly exploring and analyzing the dataset. This enabled me to quickly get access to coordinates, so I could start the study on mapping libraries. This also meant that an alternative still had to be made to automatically translate new stations. In our first stand ups we decided to continue with the mapping library study and leave the automatic translating alone. Later I came up with a solution that will be explained in the Station Configurator Extension section. In the end, I managed to very quickly generate an excel file with +-70% of stations translated. The Petrol Manager SQL server database was extended with coordinates fields, and those fields were filled from my excel.

2.3. Map study

At the time, we did not know if the internship assignment was plausible. Which is why I first had to collect information and do a study on different mapping libraries. This way we could determine if we could continue with the assignment. In my study, I came across multiple options, but most of them were paid options. And since I did not have a budget, I was asked to stick to free open-source options as much as possible. Which led me to Leaflet, Openlayers and MaplibreGL. To compare these options, I listed all their positives and negatives. Inside One Portal, I also made prototypes of these maps to better compare them. At the end we concluded that Leaflet suited our case the best.

2.4. One Portal Map Integration

The main assignment of my internship was building an interactive map. First, I focused on the basic functionalities:

- A map component to display the map container with a tile provider.
- A logical way to navigate to and open the map.
- Getting the coordinates from the database and positioning station markers on the map
- Creating a custom popup component for the markers that is injected at runtime.
- Updating the marker icon based on Alert Level.
- Integrating the map with the existing tree view. Upon selecting a network in the tree, the stations of that network appear on the map.

Once these functionalities were worked out, I had a basic functioning product. After that, I could focus more on the nice-to-have functionalities and polish

- Fullscreen mode
- Map title, map legend, tile layer switch for light and dark mode.
- A drill down button inside the popup to programmatically walk the tree to find the station and open the station details tab.
- Marker clustering to maintain visibility for densely packed areas.
- New modern station icons
- Marker bounds, zoom limits, map animations
- Caching check and refreshing the map with the rest of One Portal

With all these additions, we ended up with a cleanly designed map where clients get a visual high-level overview of the statuses of their stations. During my internship I had the opportunity to present the map in front of a large audience consisting of the whole engineering team. At the end, the feedback I received was very positive and I managed to convince them of the value this will provide for the client. In the future, if the map lands in production. It is important to monitor how the client uses the map. This will be the most important factor in telling where improvement is still possible.

2.5. Station Configurator Extension

To wrap up my assignment, the station configurator was extended to account for newly created stations and the updating of a station. When a new station is configured and inserted in the database, an automatic API call is done to Nominatim to fetch the station coordinates. It is also possible to correct any missed or wrong coordinates inside the station configurator update form. Inside the form, a separate map got added that can be toggled. This map shows the station marker and has the possibility to drag the marker around. Depending on where the marker gets dropped, that position gets sent with the form to update the station. This solution was proposed by me during stand up to correct any missed or incorrect coordinates by my Python script.

3. Personal Reflection

Looking back, this internship was a big step for me. It was the first time I moved away from clear school assignments and into a real professional environment. At first, having so much freedom was a bit of a challenge. I wasn't always sure if I was on the right track or if I was moving fast enough. However, this led me to finding a system that worked for me. Using my notebook to track every idea and question turned out to be a lifesaver. It helped me stay organized during stand-ups and weekly meetings. I learned a lot about the technical side of things, specifically working with Angular and TypeScript in a large, existing codebase. Dealing with professional coding standards was a great learning curve. But I think I grew the most in how I approach a project. Instead of just waiting for instructions, I started thinking along with the company. Proposing the Station Configurator Extension during a stand-up is a good example of that. I realized that I enjoy having control over a project and being able to make decisions that improve the final product.

The biggest hurdle was not the code but figuring out how to start when the "end goal" is still being decided. By building prototypes and doing a proper study on mapping libraries, I learned how to justify my choices with facts rather than just guessing. Presenting the map to the whole engineering team was definitely a highlight. Seeing their positive reactions made me realize that the work I did actually has value for Tokheim and their clients. Overall, this experience gave me a lot of confidence. I've proven to myself that I can take a vague idea, structure it, and turn it into a working solution. I feel much more ready to start my career as a developer now, knowing that I can handle the responsibility of an open project.

Finally, I want to mention that I really enjoyed my time at Tokheim. Beyond the technical work, the atmosphere within the team made it a great place to learn. I felt welcome from the start, and having a mentor who trusted me with this level of responsibility made the experience even more rewarding. I'm grateful for the opportunity and for the fun weeks I spent working here.